

Supplemental Material for “Canonically consistent quantum master equation”

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A. Dyson series expansion of the dynamical map - Divergences at each order

We consider a general open quantum system whose total Hamiltonian

$$H_{\text{tot}} = \underbrace{H_S + H_B}_{H_0} + \underbrace{S \otimes B}_{H_{\text{int}}}, \quad (\text{A1})$$

with H_S and S being operators acting on the system Hilbert space, and H_B and B acting on the Hilbert space of the bath. Additionally, we assume a weak interaction H_{int} (we will later refine this requirement).

Using Kubo's identity,

$$e^{\mu(A+B)} = e^{\mu A} \left(\mathbb{I} + \int_0^\mu d\nu e^{-\nu A} B e^{\nu(A+B)} \right), \quad (\text{A2})$$

we expand the total time evolution operator $U_{\text{tot}}(t, 0) = \exp[-it(H_0 + H_{\text{int}})]$ upto second order in H_{int} yielding,

$$e^{-itH_{\text{tot}}} \simeq e^{-itH_0} \left[\mathbb{I} - i \int \tilde{H}_{\text{int}}(t_1) - \iint^{t_1} \tilde{H}_{\text{int}}(t_1) \tilde{H}_{\text{int}}(t_2) \right]. \quad (\text{A3})$$

Above the symbol $\int := \int_0^t dt_1$ and $\iint^x := \int_0^t dt_1 \int_0^x dt_2$. The operator $\tilde{H}_{\text{int}}(\tau) = \exp[iH_0\tau]H_{\text{int}}\exp[-iH_0\tau]$ is defined in the interaction picture and throughout we will set $\hbar = k_B = 1$. Thus, the total density matrix

$$\varrho_{\text{tot}}(t) = e^{-it(H_0+H_{\text{int}})}\varrho_{\text{tot}}(0)e^{it(H_0+H_{\text{int}})}. \quad (\text{A4})$$

transforms as,

$$\begin{aligned} e^{itH_0}\varrho_{\text{tot}}(t)e^{-itH_0} &\simeq \varrho_{\text{tot}}(0) - i \int [\tilde{H}_{\text{int}}(t_1), \varrho_{\text{tot}}(0)] \\ &- \iint^{t_1} \left(\tilde{H}_{\text{int}}(t_1) \tilde{H}_{\text{int}}(t_2) \varrho_{\text{tot}}(0) + \text{H.c.} \right) \\ &+ \iint^t \tilde{H}_{\text{int}}(t_1) \varrho_{\text{tot}}(0) \tilde{H}_{\text{int}}(t_2). \end{aligned} \quad (\text{A5})$$

The above is the Dyson expansion up to second order in H_{int} . An alternative derivation of this expression can be obtained from a Magnus expansion up to second order in the interaction picture, followed by an expansion of the involved exponential up to second order.

In the following it is convenient to work in the interaction picture. To obtain the dynamical map $\tilde{\varrho}(t) = \tilde{\Lambda}_t[\tilde{\varrho}(0)]$ for the reduced dynamics $\tilde{\varrho}(t) = \text{Tr}_B(\varrho_{\text{tot}}(t))$, we assume factorized initial conditions $\tilde{\varrho}_{\text{tot}}(0) = \tilde{\varrho}(0) \otimes \tilde{\varrho}_B(0)$ and trace over the bath degrees of freedom to find

$$\begin{aligned} \tilde{\varrho}(t) &\simeq \tilde{\varrho}(0) - i \int \left[\tilde{S}(t_1) \langle \tilde{B}(t_1) \rangle, \tilde{\varrho}(0) \right] \\ &+ \iint^{t_1} \left(\left[\tilde{S}(t_2) \tilde{\varrho}(0), \tilde{S}(t_1) \right] \langle \tilde{B}(t_1) \tilde{B}(t_2) \rangle + \text{H.c.} \right). \end{aligned} \quad (\text{A6})$$

Herein $\langle \cdot \rangle = \text{Tr}_B(\cdot \varrho_B(0))$ denotes the bath average in the unperturbed (or “free”) evolution of the bath. Further, we assume that the bath is stationary, therefore the bath correlation function $C(t_1 - t_2) = \langle \tilde{B}(t_1) \tilde{B}(t_2) \rangle$ will only depend on the time difference and not the absolute times t_1, t_2 . Also, as in the usual convention of Born-Markov, the bath operators are chosen such that $\langle \tilde{B}(t_1) \rangle = 0$ (which can always be achieved by a redefinition of the B operators and H_S). Now by transforming $t_2 \rightarrow t_1 - t_2$ in Eq. (A6), one obtains

$$\tilde{\varrho}(t) \simeq \tilde{\varrho}(0) + \int e^{it_1 H_S} \mathcal{R}_{t_1} [e^{-it_1 H_S} \tilde{\varrho}(0) e^{it_1 H_S}] e^{-it_1 H_S}, \quad (\text{A7})$$

with $\mathcal{R}_{t_1}[\cdot] = \int_0^{t_1} dt_2 [\tilde{S}(-t_2) \cdot, S] C(t_2) + \text{H.c.}$ being the Redfield superoperator, which is the starting point for the derivation of a time local quantum master equation discussed in the main text.

From Eq. (A7) one identifies the dynamical map as $\tilde{\Lambda}_t[\cdot] \simeq \mathbb{I} + \int_0^t dt_1 \Lambda_{-t_1}^0 [\mathcal{R}_{t_1}[\Lambda_{t_1}^0[\cdot]]]$ with $\Lambda_t^0[\cdot] = e^{-itH_S} \cdot e^{itH_S}$ being the free evolution. However problems arise for the populations since in the asymptotic limit when the Redfield superoperator $\mathcal{R}_{t_1}$ is essentially time independent, the integral diverges linearly with time. This is seen explicitly for an initial state that is diagonal in H_S for which the dynamical map reads

$$\begin{aligned} \mathcal{P}^c \tilde{\Lambda}_t \tilde{\varrho}(0) &\simeq \tilde{\varrho}(0) \\ &+ \mathcal{P}^c \left(\int (t - \tau) C(\tau) \left[\tilde{S}(-\tau) \tilde{\varrho}(0), S \right] \right) + \text{H.c.}, \end{aligned} \quad (\text{A8})$$

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which is obtained from Eq. (A6) by choosing the new integration variables $\tau = t_1 - t_2$ and $\tau' = t_1 + t_2$ and $\mathcal{P}^c = \sum_n |n\rangle\langle n| \cdot |n\rangle\langle n|$ is the projector to the populations, where $|n\rangle$ are the eigenstates of H_S . Clearly the dynamical map obtained from the Dyson series expansion is inadequate in describing the long-time dynamics as it linearly diverges with t . In fact the series expansion involves divergences in each order separately.

In this work we focus on a consistent lowest order theory. However this is straightforwardly extended to higher orders and is thus valid also in the strong coupling regime by considering more terms in the Dyson series expansion of Eq. (A3).

B. Secular Lindblad (or quantum optical) master equation

Combining Eqs. (3) and (4) of the main text, we obtain the Redfield quantum master equation

$$\partial_t \varrho(t) = -i[H_S, \varrho(t)] + \mathcal{R}_t[\varrho(t)]. \quad (\text{A9})$$

From this, a Lindblad form can be found in the ultraweak coupling limit where the system-bath coupling strength is weak when compared to all energy splittings $E_n - E_m$, $n \neq m$, in the system. By performing the standard secular (or rotating-wave) approximation, we find [1]

$$\begin{aligned} \partial_t \varrho(t) = & -i[H_S + H^{\text{LS}}, \varrho(t)] \\ & + \sum_{nm} |S_{nm}|^2 2W'_{nm} \mathcal{L}(|n\rangle\langle m|)[\varrho(t)]. \end{aligned} \quad (\text{A10})$$

with Lamb shift

$$H^{\text{LS}} = \sum_{nm} |S_{nm}|^2 W''_{nm} |m\rangle\langle m|. \quad (\text{A11})$$

This is the secular Lindblad (or quantum optical) master equation.

C. Canonical perturbation theory for the mean force Gibbs state

Following the eigenstate thermalization hypothesis, in thermal equilibrium we expect the total system-bath composite to thermalize with inverse temperature β (determined by the mean energy). By tracing out the bath degrees of freedom, the reduced state of the system is given by the mean force Gibbs state

$$\lim_{t \rightarrow \infty} \varrho(t) = \frac{\text{Tr}_B(e^{-\beta H_{\text{tot}}})}{Z_{\text{tot}}}, \quad (\text{A12})$$

with $H_{\text{tot}} = H_0 + H_{\text{int}}$ where $H_0 = H_S + H_B$ is the Hamiltonian of the system and bath respectively and H_{int} is the system-bath interaction. Note that Eq. (A12) allows for the identification of the Hamiltonian of mean force as in

Refs. [2, 3]. Here, however, we want to extract the correction to the Gibbs state perturbatively in the interaction Hamiltonian. By again making use of the Kubo identity, Eq. (A2), we find to second order,

$$\begin{aligned} e^{-\beta H_{\text{tot}}} \simeq & e^{-\beta H_0} \left[\mathbb{I} + \int_0^\beta d\mu \tilde{H}_{\text{int}}(-i\mu) \right. \\ & \left. + \int_0^\beta d\mu \int_0^\mu d\xi \tilde{H}_{\text{int}}(-i\mu) \tilde{H}_{\text{int}}(-i\xi) \right] \end{aligned} \quad (\text{A13})$$

with freely evolving interaction-picture operators in imaginary time $\tilde{H}_{\text{int}}(-i\mu) = e^{\mu H_0} H_{\text{int}} e^{-\mu H_0}$. By assuming a bath at equilibrium and by taking the partial trace only even orders of the expansion contribute to the mean force Gibbs state. For a system bath coupling Hamiltonian of the form $H_{\text{int}} = S \otimes B$ this is given by,

$$\begin{aligned} \text{Tr}_B(e^{-\beta H_{\text{tot}}}) = & e^{-\beta H_S} Z_B \left[\mathbb{I} + \right. \\ & \left. \int_0^\beta d\mu \int_0^\mu d\xi \tilde{S}(-i\mu) \tilde{S}(-i\xi) C(-i(\mu - \xi)) \right], \end{aligned} \quad (\text{A14})$$

with bath correlation $C(-i(\mu - \xi)) = \text{Tr}_B(\varrho_B \tilde{B}(-i\mu) \tilde{B}(-i\xi))$ evaluated at imaginary time. Apart from the system-bath coupling, the Caldeira Leggett model in Eq. (A33) also considers a renormalization Hamiltonian that renormalizes the frequency shift caused by the reservoir. For an Ohmic bath with Drude cutoff it takes the form $H_{\text{RN}} = \gamma \omega_D S^2 / 2$ [4] and modifies the bath correlation, $C(-i\lambda) \rightarrow C(-i\lambda) - \gamma \omega_D \delta(\lambda) / 2$. We arrive at the second order expansion of the mean force Gibbs state,

$$\begin{aligned} \frac{\text{Tr}_B(e^{-\beta H_{\text{tot}}})}{Z_{\text{tot}}} \simeq & \left(\mathbb{I} + \bar{Q} \right) \left[\frac{e^{-\beta H_S}}{Z_S} \right] \\ & - \frac{e^{-\beta H_S}}{Z_S} \text{Tr}_S \bar{Q} \left[\frac{e^{-\beta H_S}}{Z_S} \right] \end{aligned} \quad (\text{A15})$$

where we define the action of the superoperator

$$\begin{aligned} \bar{Q} \left[\frac{e^{-\beta H_S}}{Z_S} \right] = & \frac{e^{-\beta H_S}}{Z_S} \int_0^\beta d\mu \int_0^\mu d\xi \tilde{S}(-i\mu) \tilde{S}(-i\xi) C(-i(\mu - \xi)) \end{aligned} \quad (\text{A16})$$

on the canonical Gibbs state. The last term in Eq. (A15) that is quadratic in the canonical Gibbs state is a consequence of the normalization. In the dynamical theory for which we propose a canonically consistent master equation, which by construction guarantees trace preservation, this non-linear term may be dropped.

In the following we explicitly calculate the action of the operator above and express it in terms of the Redfield dissipator. By rotating the coordinate system and introducing the independent integration variables $s \equiv \mu + \xi$

and $u \equiv \mu - \xi$ we obtain

$$\bar{Q} \left[\frac{e^{-\beta H_S}}{Z_S} \right] = \frac{e^{-\beta H_S}}{Z_S} \int_0^\beta du C(-iu) \int_{u/2}^{\beta-u/2} ds e^{sH_S} \tilde{S}(-iu/2) \tilde{S}(iu/2) e^{-sH_S}, \quad (\text{A17})$$

in which we can solve the s -integration independently. The integral is qualitatively different for either coherences and populations. To keep the notation simple we

introduce the projection operators

$$\mathcal{P} = \sum_{n \neq m} |n\rangle\langle n| \cdot |m\rangle\langle m|, \quad (\text{A18})$$

which projects to the coherent subspace where $\Delta = [H_S, \cdot]$ is invertible, and the complementary space

$$\mathcal{P}^c = \sum_n |n\rangle\langle n| \cdot |n\rangle\langle n|, \quad (\text{A19})$$

for the populations. In both subspaces the s -integral can be evaluated as,

$$\mathcal{P} \int ds e^{sH_S} \cdot e^{-sH_S} = \mathcal{P} \frac{1}{\Delta} e^{sH_S} \cdot e^{-sH_S}, \quad (\text{A20})$$

$$\mathcal{P}^c \int ds e^{sH} \cdot e^{-sH} = \mathcal{P}^c s. \quad (\text{A21})$$

Thus, we obtain the simplified action of the $\bar{Q}$ superoperator as,

$$\begin{aligned} \bar{Q} \left[\frac{e^{-\beta H_S}}{Z_S} \right] = & \mathcal{P} \frac{1}{\Delta} \left[S \left(\int_0^\beta du C(-iu) e^{-uH_S} S e^{uH_S} \right) \frac{e^{-\beta H_S}}{Z_S} - \frac{e^{-\beta H_S}}{Z_S} \left(\int_0^\beta du C(-iu) e^{-uH_S} S e^{uH_S} \right)^\dagger S \right] \\ & + \mathcal{P}^c \frac{e^{-\beta H_S}}{Z_S} S \int_0^\beta du C(-iu) (\beta - u) e^{-uH_S} S e^{uH_S}. \end{aligned} \quad (\text{A22})$$

Integrals of the form above are well known from the Redfield theory. In fact it turns out that the first line can be rewritten with the Redfield superoperator. To this end we make use of the identity [4],

$$-\int_0^\beta du C(-iu) e^{-uE} = W''(E) + e^{-\beta E} W''(-E) \quad (\text{A23})$$

with $W''(x)$ being the imaginary part of the Fourier-Laplace transform of the bath correlation function,

$$W(E) = \int_0^\infty dt C(t) e^{-iEt}. \quad (\text{A24})$$

The real part of the above integral is typically referred to as the rates whereas the imaginary part is known as the Lamb shift. With this we can rewrite the second order contribution as

$$\begin{aligned} \bar{Q} \left[\frac{e^{-\beta H_S}}{Z_S} \right] = & \mathcal{P} \frac{1}{i\Delta} \mathcal{R}_\infty \left[\frac{e^{-\beta H_S}}{Z_S} \right] \\ & + \mathcal{P}^c \frac{e^{-\beta H_S}}{Z_S} S \int_0^\beta du C(-iu) (\beta - u) e^{-uH_S} S e^{uH_S}, \end{aligned} \quad (\text{A25})$$

with the static Redfield superoperator

$$\mathcal{R}_\infty[\varrho] = -S\mathbb{S}\varrho + S\varrho\mathbb{S}^\dagger - \varrho\mathbb{S}^\dagger S + \mathbb{S}\varrho S. \quad (\text{A26})$$

and the convolution operator

$$\mathbb{S} = \int_0^\infty dt C(t) e^{-iH_S t} S e^{iH_S t}. \quad (\text{A27})$$

When $\mathbb{S}$ is expressed in the eigenbasis of H_S it simplifies as $\langle n|\mathbb{S}|m\rangle = S_{nm}W(\Delta_{nm})$ with $S_{nm} = \langle n|S|m\rangle$ and eigenenergy difference $\Delta_{nm} = E_n - E_m$. Note, the identity Eq. (A23) contains only the imaginary parts of the Fourier-Laplace transform of the bath correlator. However, in Eq. (A25) we took the full Redfield superoperator including the real part, i.e., the Redfield rates. Since the Redfield superoperator in Eq. (A25) acts on the canonical Gibbs state, the additional parts due to the rates sum upto zero as they follow detailed balance. In other words, Eqs. (A22) and (A25) are equal to each other as a consequence of the canonical Gibbs state obeying detailed balance.

We also want to infer the superoperator in the orthogonal subspace for the populations. These terms arise in

the expansion above as integrals of the form

$$-\int_0^\beta du C(-iu) u e^{-ux} = \frac{\partial}{\partial x} (W''(x) + e^{-x\beta} W''(-x)). \quad (\text{A28})$$

By making use of this identity in the eigenbasis of H_S the populations for the second order contribution are given by [4]

$$\langle n | \bar{Q} \left[\frac{e^{-\beta H_S}}{Z_S} \right] | n \rangle = \sum_l |S_{nl}|^2 \left[V''(\Delta_{nl}) \frac{e^{-\beta E_l}}{Z_S} - V''(\Delta_{ln}) \frac{e^{-\beta E_n}}{Z_S} + W''(\Delta_{ln}) \frac{\partial}{\partial E_n} \frac{e^{-\beta E_n}}{Z_S} \right]. \quad (\text{A29})$$

Here the first two terms describe a Pauli rate equation that can be represented by a Lindblad equation with rates given by the derivatives of the Redfield Lamb shifts $V''(x) = \partial W''(x)/\partial x$. In other words, the Lamb shifts that are typically ignored play an important role as they appear as rates at a higher order. The last term describes yet another contribution that acts as a derivative onto the state itself. Instead of writing the derivative only for the equilibrium state as $\partial/\partial E_n e^{-\beta E_n}/Z_S = -\beta e^{-\beta E_n}/Z_S$ we find a much better performance of the CCQME if we infer the derivative operator as in reference [4]. To do so we formally solve the Pauli rate equation for the zeroth

order steady state populations $p_n = e^{-\beta E_n}/Z_S$

$$p_n = \frac{\sum_{l \neq n} |S_{nl}|^2 W'(\Delta_{nl}) p_l}{\sum_{l \neq n} |S_{ln}|^2 W'(\Delta_{ln})}, \quad (\text{A30})$$

and take the derivative on both sides to find

$$\frac{\partial p_n}{\partial E_n} = \frac{\sum_{l \neq n} |S_{nl}|^2 [V'(\Delta_{nl}) p_l + V'(\Delta_{ln}) p_n]}{\sum_{l \neq n} |S_{ln}|^2 W'(\Delta_{ln})} \quad (\text{A31})$$

with $V'(x) = \partial W'(x)/\partial x$. To conclude this section we identify the superoperator $\bar{Q}$ for the CCQME from the mean force Gibbs state that is easily obtained from the Redfield rates/Lamb shifts and their derivatives according to

$$\bar{Q}[\varrho] = \mathcal{P} \frac{1}{i\Delta} \mathcal{R}_\infty[\varrho] + \mathcal{P}^c \left(\sum_{nl} |S_{nl}|^2 \left[V''(\Delta_{nl}) \mathcal{L}(|n\rangle\langle l|)[\varrho] + W''(\Delta_{ln}) |n\rangle \frac{\partial}{\partial E_n} \langle n|\varrho|n\rangle \langle n| \right] \right) \quad (\text{A32})$$

with Lindblad operator $\mathcal{L}(L) = L \cdot L^\dagger - \frac{1}{2}\{L^\dagger L, \cdot\}$ and the formal derivative Eq. (A31) for the populations.

We may further increase accuracy by going beyond lowest order perturbation theory and taking into account higher order corrections to the mean force Gibbs state in Eq. (A15).

D. Exact master equation for damped harmonic oscillator

In order to corroborate our approach, we test our method against the exact solution of the damped harmonic oscillator, whose position is coupled to a continuum of oscillator modes. The total system-bath Hamil-

tonian is given by the Caldeira Leggett model [5],

$$H_{\text{tot}} = \frac{p^2}{2M} + \frac{M\Omega^2}{2} q^2 + \sum_k \left[\frac{p_k^2}{2m_k} + \frac{m_k \omega_k^2}{2} \left(q_k - \frac{c_k}{m_k \omega_k^2} q \right)^2 \right], \quad (\text{A33})$$

with position q (q_k) and momentum p (p_k) of the central (bath) oscillator(s). In the following we work in units where we set the central oscillator mass to one, i.e. $M = 1$. This model has been extensively studied in references [6–8]. Here we are interested in an effective description of the central oscillator that is exact and does not rely on perturbative expansions in system-bath coupling. For this we assume an initial product state of system and bath and that the bath initially is in thermal equilibrium with inverse temperature β . Under this conditions it has been shown [8] that the dynamics of the central oscillator follows the exact master equation

$$\partial_t \varrho_{\text{ex}} = -i \left[\frac{p^2}{2} + \frac{\gamma_q(t)}{2} q^2, \varrho_{\text{ex}} \right] - D_p(t)[q, [q, \varrho_{\text{ex}}]] - \frac{i}{2} \gamma_p(t)[q, \{p, \varrho_{\text{ex}}\}] + D_q(t)[q, [p, \varrho_{\text{ex}}]]. \quad (\text{A34})$$

Above $\gamma_q(t)$, $\gamma_p(t)$, $D_p(t)$ and $D_q(t)$ are time-dependent parameters that depend on the bath properties and details of the coupling. It is a special property of the damped harmonic oscillator that all non-Markovian features are fully encoded in the time-dependence of the parameters without the need for non-local integral kernels in the master equation [9, 10]. It is easy to check that Eq. (A34) is hermiticity and trace preserving. However, it should be noted that it is not a Lindblad master equation even in the Markovian limit for which the time-dependent parameters are replaced by their asymptotic values. The necessity for completely positive maps is strongly debated in the community and any physical quantum map should only guarantee the weaker condition of positivity [11]. This is indeed the case for the above master equation (A34).

The derivation of the above exact master equation and the details of the parameters are discussed in Ref. [8]. Essentially there are two building blocks for the parameters. On the one hand there is the system's autocorrelation $G(t) = i \langle [q(t), q(0)] \rangle$ that satisfies the damped oscillator equation

$$\ddot{G}(t) + \int_0^t \kappa(t-\tau) \dot{G}(\tau) d\tau + \Omega^2 G(t) = 0 \quad (\text{A35})$$

with initial conditions $G(0) = 0$ and $\dot{G}(0) = 1$. The damping kernel $\kappa(\tau) = \sum_k (c_k^2 / m_k \omega_k^2) \cos(\omega_k \tau) =$

$\int_0^\infty (J(\omega)/\omega) \cos(\omega t) d\omega$ is determined by the bath parameters. The expression in terms of the spectral density $J(\omega)$ holds in the continuum limit of the bath. In our work we consider an Ohmic spectral density with Drude cutoff

$$J(\omega) = \frac{\gamma \omega}{1 + (\omega/\omega_D)^2}, \quad (\text{A36})$$

for which one finds an analytic solution of the autocorrelation in Laplace space [8],

$$\begin{aligned} \tilde{G}(z) &= \frac{1}{z^2 + z\tilde{\kappa}(z) + \Omega^2} \\ &= \frac{\omega_D + z}{z^3 + z^2\omega_D + z(\Omega^2 + \gamma\omega_D) + \omega_D\Omega^2}, \end{aligned} \quad (\text{A37})$$

where the second line follows explicitly for the Drude bath. By inverse Laplace transformation two of the four parameters are readily obtained,

$$\gamma_q(t) = \frac{\ddot{G}(t)^2 - \dot{G}(t)\ddot{G}(t)}{\dot{G}(t)^2 - G(t)\ddot{G}(t)}, \quad (\text{A38})$$

$$\gamma_p(t) = \frac{G(t)\ddot{G}(t) - \dot{G}(t)\dot{G}(t)}{\dot{G}(t)^2 - G(t)\ddot{G}(t)}, \quad (\text{A39})$$

and are particularly temperature independent. For notational simplicity we use the dots above the G to indicate the order of the time derivative.

On the other hand there is the bath's autocorrelation function $C(t) = \langle B(t)B(0) \rangle$ for the bath operator $B = \sum_k -c_k q_k$ that is well known from the Redfield formalism [4]. The diffusive parameters are given by

$$D_q(t) = \frac{1}{2} \ddot{K}_q(t) - K_p(t) + \gamma_q(t) K_q(t) + \frac{\gamma_p(t)}{2} \dot{K}_q(t), \quad (\text{A40})$$

$$D_p(t) = \frac{1}{2} \dot{K}_p(t) + \frac{\gamma_q(t)}{2} \dot{K}_q(t) + \gamma_p(t) K_p(t). \quad (\text{A41})$$

Here $K_q(t)$ and $K_p(t)$ are real time influence kernels [6, 8] given by,

$$K_q(t) = \int_0^t ds \int_0^t du \text{Re}[C(|s-u|)] G(s) G(u), \quad (\text{A42})$$

$$K_p(t) = \int_0^t ds \int_0^t du \text{Re}[C(|s-u|)] \dot{G}(s) \dot{G}(u). \quad (\text{A43})$$

In this work we use the exact result to benchmark the canonically consistent quantum master equation (CCQME) derived in the main text. For the numerics we solve the master equations in the Markovian limit that is

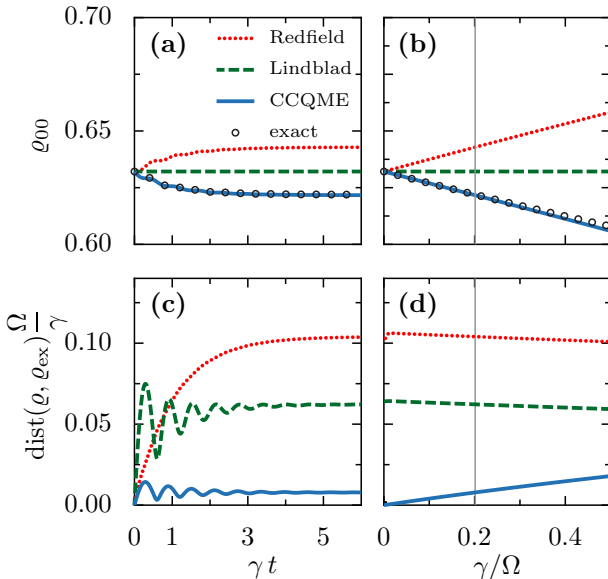


FIG. 1. Analogue to Fig. 1 in the main text except for a higher bath temperature $T/\Omega = 1$.

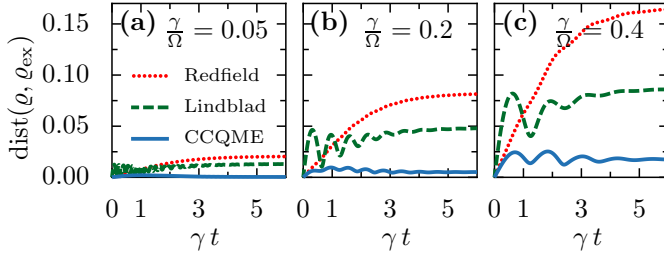


FIG. 2. Error for Redfield (dotted red), Lindblad (dashed green), and CCQME (solid blue) calculated as trace distance to exact result as a function of scaled time for different values of the coupling strength (a) $\gamma/\Omega = 0.05$, (b) $\gamma/\Omega = 0.2$, and (c) $\gamma/\Omega = 0.4$. The initial state is $\varrho(0) = \exp[-H_S/\Omega]/Z_S$ and the bath temperature is $T/\Omega = 0.3$. For the simulation we take the Markovian master equations with Drude cutoff $\omega_D/\Omega = 5$.

for times where the parameters $\gamma_q(t)$, $\gamma_p(t)$, $D_q(t)$, $D_p(t)$ and the Redfield rates/Lamb shifts are relaxed to their asymptotic values.

Similar to the analysis in the main text we calculate the ground state population in Fig. 1(a), (b) and in Fig. 1(c), (d) the trace distance to the exact result,

$$\text{dist}(\varrho, \varrho_{\text{ex}}) = \frac{1}{2} \text{Tr}(\sqrt{(\varrho(t) - \varrho_{\text{ex}}(t))^2}). \quad (\text{A44})$$

Here we consider a larger bath temperature $T/\Omega = 1$ for which quantum effects are less relevant. Compared to the system time scale given by $1/\Omega$, in this regime the bath correlation function decays much faster and the Markov approximation is well justified. In this high-temperature regime the Redfield and Lindblad equations are usually expected to be valid approximations. This is shown in Fig. 1 where the results are close together and in particular the Redfield solution remains positive. However, strictly speaking also here both the Redfield and Lindblad result only remain valid for ultraweak coupling, whereas the CCQME accurately captures the ground state population also for finite coupling in Fig. 1 and overall has the smallest trace distance to the exact result [Figs. 1 and 2]. Contrastingly the Lindblad steady state predictions are independent of the coupling strength and the Redfield result increases with coupling strength as seen in Fig. 1(b) and Fig. 1(b) [main text]. As described in the main text, for stronger coupling the effective temperature should increase and the higher lying states should get populated, a property only captured by the CCQME.

In Fig. 2 we plot the trace distance for different values of the system-bath coupling strength γ . By construction the steady state of the CCQME accurately captures finite coupling effects but it is not obvious that it also holds for the dynamics. In the main text we analyze the time averaged trace distance as a function of bath temperature and coupling strength. Here we give an overview for the dynamics of the trace distance for different values of the coupling strength. All panels

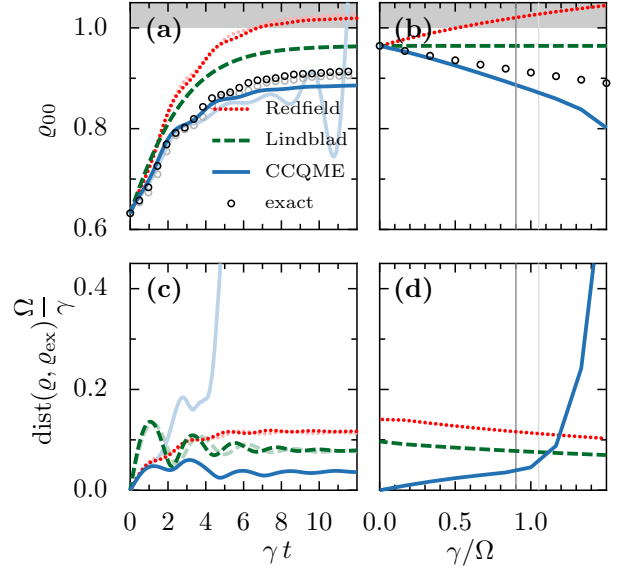


FIG. 3. Analogue to Fig. 1 of the main text except for $\gamma/\Omega = 0.9$ (dark lines and symbols), $\gamma/\Omega = 1.01$ (light lines and symbols), and $\omega_D/\Omega = 1$.

show the same results qualitatively, with the CCQME outperforming the other QMEs for the error. Moreover, for the CCQME the trace distance decreases monotonically with smaller values of the coupling not only in the steady-state regime but also throughout the evolution.

E. CCQME regime of validity

For weak but finite coupling, i.e. $\gamma/\Omega \leq 0.5$ the CCQME very accurately agrees with the exact dynamics (see main text). That is because the CCQME is a second-order consistent master equation whose coupling strength γ is a perturbative (control) parameter, that governs the precision of the simulated dynamics. If we increase γ beyond the second order coupling regime, i.e. $\gamma/\Omega = 0.9$, the CCQME steady state starts to deviate significantly from the exact result but the solution is still positive [see dark lines and symbols in Fig. 3]. For strong coupling $\gamma/\Omega > 1$ the approximations made for the CCQME are no longer valid and the solution may violate positivity [see light lines and symbols in Fig. 3]. On the other hand, for weak-coupling approaches like Redfield, positivity is already violated for $\gamma/\Omega \approx 0.5$, thereby doubling the validity range.

The main idea of the CCQME is to combine a dynamical theory with the exact mean force Gibbs state. Specifically, we correct the Born approximation that assumes that the total density matrix of the system-bath composite is in the tensor product form at all times, i.e., $\varrho_{\text{tot}} = \varrho(t) \otimes \rho_B$. Even at higher orders the Born approximation is unavoidable [12–14]. Thus, using the same line of thought in the main text one can use the general

expansion of the mean force Gibbs state to replace the freely evolving state $\Lambda_0^t[\varrho(0)]$, so as to obtain the generalized higher order CCQME [15].

F. CCQME for a system connected to multiple baths

In this section we implement the CCQME to a minimal transport setup wherein the harmonic oscillator system is driven by two independent thermal baths as depicted schematically in Fig. 4(a). For uncorrelated baths the damping kernel and bath correlation functions simply add up such that the autocorrelation function Eq. (A37) and the real time influence kernels Eqs. (A42) and (A43) are altered to

$$\tilde{G}(z) = \frac{1}{z^2 + z \sum_i \tilde{\kappa}^i(z) + \Omega^2}, \quad (\text{A45})$$

$$K_q(t) = \int_0^t ds \int_0^t du \sum_i \text{Re}[C^i(|s-u|)] G(s) G(u), \quad (\text{A46})$$

$$K_p(t) = \int_0^t ds \int_0^t du \sum_i \text{Re}[C^i(|s-u|)] \dot{G}(s) \dot{G}(u). \quad (\text{A47})$$

The remaining structure of equations (A39)-(A41) remain the same allowing us to simulate the exact quantum master equation Eq. (A34) for the nonequilibrium scenario.

Following the line of arguments of the main text we start with the Redfield equation

$$\partial_t \varrho(t) = -i[H_S, \varrho(t)] + (\mathcal{R}_\infty^l + \mathcal{R}_\infty^r)[\Lambda_t^0[\varrho(0)]], \quad (\text{A48})$$

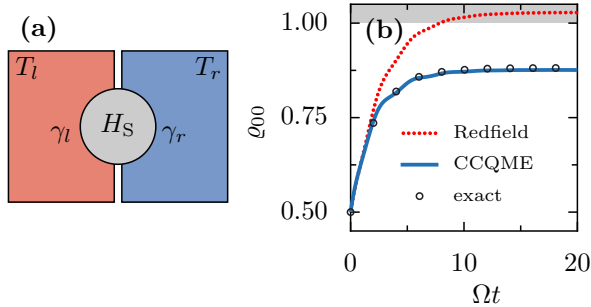


FIG. 4. Harmonic oscillator coupled to two independent baths illustrated in panel (a). Dynamics for the ground state population when the system is initiated in a superposition of the ground and first excited state $|\psi(0)\rangle = (1/\sqrt{2})(|0\rangle + |1\rangle)$ under the Redfield equation (dotted red), boundary driven CCQME (solid blue), and the exact solution (open black circles) in panel (b). For the simulation $N = 20$ first levels are taken. The bath parameters are $\gamma_l/\Omega = \gamma_r/\Omega = 0.2$, $T_l/\Omega = 0.3$, $T_r/\Omega = 0.5$, and $\omega_D/\Omega = 5$.

where the Redfield operators for the left and right bath respectively act on the initial state. Replacing the initial state according to the inverse map,

$$\Lambda_t^0[\varrho(0)] \simeq (\mathbb{I} - \bar{Q}^l - \bar{Q}^r)[\varrho(t)], \quad (\text{A49})$$

wherein the superoperators are obtained from Eqs. (A31) and (A32) we arrive at the boundary driven CCQME,

$$\partial_t \varrho(t) = -i[H_S, \varrho(t)] + (\mathcal{R}_\infty^l + \mathcal{R}_\infty^r)[(\mathbb{I} - \bar{Q}^l - \bar{Q}^r)[\varrho(t)]]. \quad (\text{A50})$$

It's important to note here that even though the left (l) and right (r) baths are uncorrelated the dissipators beyond weak coupling (second order) are not. This property can also be observed in the exact formulation or fourth-order QMEs [12, 16] and is simple to note in our boundary driven CCQME.

For Ohmic baths with a Drude cutoff the dynamics for the ground state population is depicted in Fig. 4(b). As expected for low temperatures and finite coupling the Redfield theory shows an unphysical behaviour with the ground state population exceeding one. On the other hand, the CCQME shows a surprisingly good agreement with the exact solution demonstrating that our approach is easily extendable to study the dynamics of boundary driven quantum systems. In future works we will connect it with recent advances in evaluating the asymptotic state of nonequilibrium quantum systems driven by multiple reservoirs [17] or an external drive [18, 19].

G. Spin-boson model

As compared to other numerically exact approaches such as the hierarchy equation of motion (HEOM) approach [20, 21], the CCQME does not require more resources as compared to the Redfield formalism. Therefore it can be applied to larger systems and for low bath temperatures at which the resources for the HEOM approach scale exponentially.

In this section we compare the CCQME with the HEOM for the spin-boson model. The spin-boson model can also be solved using the reaction-coordinate approach which is less computationally expensive than HEOM but still more expensive than the CCQME [22-24]. The total system-bath Hamiltonian reads as

$$H_{\text{tot}} = \frac{\varepsilon}{2} \sigma_x + \sigma_z \sum_k g_k (a_k^\dagger + a_k) + \sum_k \omega_k a_k^\dagger a_k \quad (\text{A51})$$

and we consider an Ohmic spectral density with exponential cutoff

$$J(\omega) = \sum_k g_k^2 \delta(\omega - \omega_k) = \lambda(\omega/\omega_c) e^{-\omega^2/\omega_c^2}. \quad (\text{A52})$$

For the Redfield dynamics we take the time dependent rates $W(E, t) = \int_0^t C(\tau) e^{-iE\tau} d\tau$ with bath correlation

function given by

$$C(\tau) = \int_0^\infty J(\omega) [\coth(\beta\omega/2) \cos(\omega\tau) - i \sin(\omega\tau)] d\omega. \quad (\text{A53})$$

Here for ultraweak coupling and high bath temperature, the Redfield equation guarantees positivity as shown explicitly for the ground state population in Fig. 5. However it does not match the numerically exact HEOM dynamics. For the CCQME we combine the time dependent Redfield equation with the static Q -tensor (see main text) and obtain accurate dynamics without the computational complexity of HEOM.

H. Ising chain

Having benchmarked the performance for a single-spin-boson model, here we demonstrate the performance of the CCQME for a genuine many-body system, where we choose an Ising chain of length L with open boundary conditions and external fields in x - and z -directions coupled to a bosonic Lorentz-Drude bath. The corresponding total system-bath Hamiltonian reads,

$$H_{\text{tot}} = -J \sum_{i=1}^{L-1} \sigma_z^i \sigma_z^{i+1} + \sum_{i=1}^L (h_x^i \sigma_x^i + h_z^i \sigma_z^i) + \sigma_z^{L/2} \sum_k g_k (a_k^\dagger + a_k) + \sum_k \omega_k a_k^\dagger a_k, \quad (\text{A54})$$

where the bath is coupled locally via a dephasing coupling to the central site. In order to avoid near degeneracies in

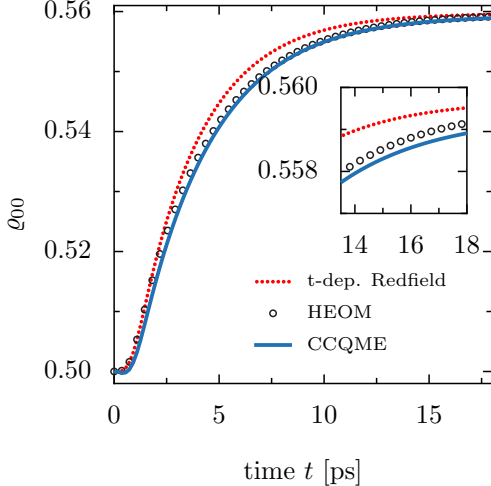


FIG. 5. Comparison of the spin-boson ground-state dynamics with the HEOM data taken from [21] (open circles), time-dependent Redfield (dotted red), and the CCQME result (solid blue). The parameters are chosen as $\varepsilon = \pi/2 \text{ ps}^{-1}$, $T = 50 \text{ K}$, $\lambda = 0.01485 \text{ ps}^{-1}$, and $\omega_c = 2.2 \text{ ps}^{-1}$.

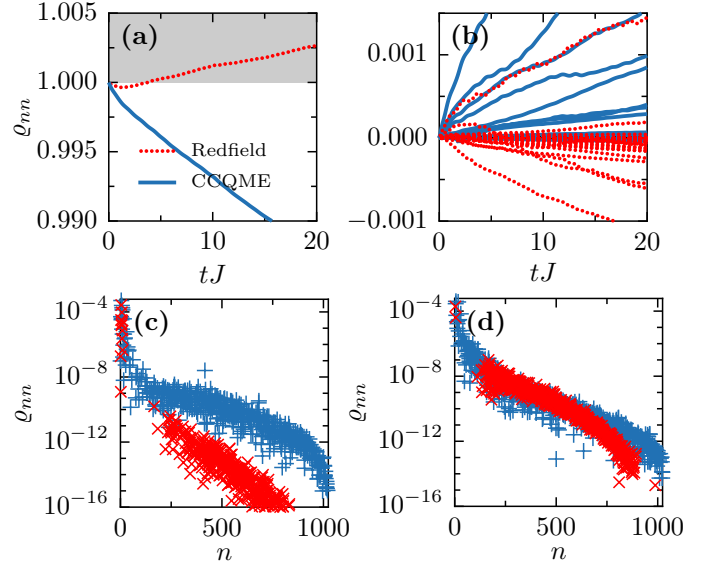


FIG. 6. Many-body populations of a spin chain of length $L = 10$ coupled locally to a Lorentz-Drude bath for (a) the many-body ground state and (b) all excited states as a function of time t . Snapshot of the populations at (c) $t = 1/J$ and (d) $t = 20/J$, red (blue) crosses depict Redfield (CCQME). On the logarithmic scale, only positive populations can be shown. Redfield eventually predicts negative populations that are of the order of the coupling strength, while the CCQME remains mostly positive at long times. The parameters are chosen as $\gamma = 0.04J$, $T = 0.9J$, and $\omega_D = 5J$.

the many-body spectrum we choose slowly twisting fields according to

$$h_x^i = 0.8J + \frac{i-1}{L-1} 0.2J, \quad (\text{A55})$$

$$h_z^i = 0.7J - \frac{i-1}{L-1} 0.2J, \quad (\text{A56})$$

that breaks the underlying symmetries lifting degeneracies.

We consider a Lorentz-Drude spectral density

$$J(\omega) = \sum_k g_k^2 \delta(\omega - \omega_k) = \frac{\gamma\omega}{1 + (\omega/\omega_D)^2}, \quad (\text{A57})$$

with coupling strength γ and cut-off ω_D . Such a spectral density is not easy to treat via HEOM [21] or pseudomode mapping [23, 25–27] (also known as mesoscopic leads approach). Hence the relatively large system size of $L = 10$ (and Hilbert space dimension $2^{10} = 1024$) is unaccessible for those methods using a machine with 64GB RAM that we use, while we are still able to solve the CCQME.

In Fig. 6, we plot the dynamics of the populations Q_{nn} of the many-body eigenstates for a chain of length $L = 10$ both for the Redfield master equation as well as for the CCQME, starting in the pure many-body ground state $\varrho(0) = |\psi_0\rangle\langle\psi_0|$. Remarkably, even though the Redfield solution is unphysical, with many negative populations

of the excited states [Fig. 6(b)] and a ground-state population above unity [Fig. 6(a)], the ground-state population for the CCQME solution is physical at all times, and positivity violation of the excited state populations is rare and of small magnitude. At $t = 20/J$ the negative populations in Redfield sum up to $\sum_{n, \varrho_{nn} < 0} \varrho_{nn}(t) \approx -4.3 \cdot 10^{-2}$ which is of the order of the coupling strength

$\gamma/J = 4 \cdot 10^{-2}$. For the CCQME the negative populations only arise on second order in the coupling strength $\sum_{n, \varrho_{nn} < 0} \varrho_{nn} \approx -6.7 \cdot 10^{-5}$ which is an advantage by three orders of magnitude. The long-time state of the CCQME in Fig. 6(d) is also consistent with ETH with populations being a smooth function of energy and small fluctuations around it.

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